

NETHEESWARAN A – DATA SCIENTIST

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PROFESSIONAL SUMMARY

Data Scientist with production experience delivering LLM/RAG and classic ML systems end to end, covering data pipelines, feature engineering, training, serving, and CI/CD on AWS, Azure, and GCP. Skilled with LangChain, Hugging Face Transformers, Pinecone, prompt engineering, and retrieval evaluation. Built containerized services with Docker and Kubernetes and deployed with MLflow/Kubeflow, improving latency, reliability, and model quality in real use.

SKILLS

Languages & data	: Python, R, SAS, SQL, PL/SQL, PySpark, Pandas, NumPy
ML/AI & Analytics	: scikit-learn, TensorFlow, PyTorch, MLflow, DVC, NLP, LLMs, Hugging Face, LangChain, OpenAI API, Matplotlib, Seaborn
Platforms & Databases	: Snowflake, Databricks, MySQL, MongoDB, Oracle, Pinecone (Vector DB)
Cloud, MLOps & Serving	: AWS (EC2, S3, Glue, Redshift, Lambda, Bedrock, SageMaker), Azure (Data Lake, Data Factory, Synapse), GCP (BigQuery, Cloud Storage), Docker, Git, GitHub Actions, Kubernetes, BentoML, Flask, FastAPI, Streamlit, Jupyter, VS Code, Jira

EXPERIENCE

Data & ML Engineer – TATA Consultancy Services, Bengaluru

Mar 2022 – May 2024

- Analyzed raw financial data quality and performed data cleansing, designing structured data models (Snowflake, star schema, SCD2) for scalable risk analytics and KPI measurement using advanced SQL and Python.
- Developed robust ETL/ELT pipelines using Python, SQL, Azure Data Factory, Airflow, and Databricks to integrate high-volume data sources, reducing reporting cycles by 40%.
- Performed comprehensive EDA on large volumes of financial datasets using Python (Pandas, NumPy, PySpark) and SQL, informing feature engineering for machine learning models.
- Applied predictive modeling, machine learning, clustering and classification techniques including regression, time-series, and outlier detection to generate actionable business insights.
- Designed accurate prediction algorithms using advanced statistical modeling workflows, supporting stress testing and business improvements through quantitative analytics.
- Built interactive dashboards and data visualizations in Tableau and Power BI to communicate complex quantitative analysis and model outputs to business stakeholders.
- Collaborated with engineering teams to deploy machine learning models to production as APIs (FastAPI/Flask) on Docker/Kubernetes with CI/CD workflows.
- Mentored junior analysts in Python, SQL, Git workflows, and BI tools, fostering best practices in data quality and reproducible ML development.

Snowflake Cloud Analyst – TATA Consultancy Services, Bengaluru

Dec 2020 – Mar 2022

- Led Azure Teradata to Snowflake migration validation (12TB, 420+ tables), applying reconciliation frameworks to ensure accurate datasets for downstream machine learning systems using advanced SQL.
- Built an automated validation suite of 680 SQL/Python tests (pytest, ADF, GitHub Actions, Slack/email alerts), improving defect detection by 97% and reducing QA cycle time by 48 hours per release.
- Designed Tableau monitoring dashboard for data quality KPIs, improving transparency and delivering actionable insights for business stakeholders through data visualization.
- Authored comprehensive documentation and governance playbooks for quantitative analytics workflows, reducing MTTR by 35% and strengthening audit readiness.

PROJECTS

AI Medical Chatbot - Python · OpenAI · LangChain · Pinecone - [Code](#)

Built a RAG chatbot that answers clinician queries from PDFs with source citations and improved top-k retrieval hit rate about 12 pp and reduced hallucinations via chunking and hybrid search. Median Latency is about 1.2s with async chains and the groundedness pass rate is 88% on a small gold set with logs and evaluations tracked in MLflow. Deployed on EC2 with GitHub Actions CI/CD using Docker with health checks and rollback, delivering 99.9% measured uptime.

Pneumonia Detection from X-rays - PyTorch · CNN · Streamlit · AWS - [Code](#)

Designed a VGG16-based deep learning pipeline leveraging transfer learning and class-imbalance handling to identify pneumonia from chest X-rays with high precision (ROC-AUC $\approx$ 0.96, F1 $\approx$ 0.92, recall $\approx$ 0.94 on a held-out test set), enabling real-time predictions to support early screening in healthcare environments.

Shoulder X-ray Implant Classification – TensorFlow · EfficientNet · CNN · Jupyter Notebook - [Code](#)

Developed an EfficientNet-B1 deep learning pipeline leveraging transfer learning and class-balancing techniques to classify shoulder X-ray implants across 16 models from 4 manufacturers with high accuracy (Accuracy $\approx$ 0.88, Macro Avg $\approx$ 0.78, Weighted Avg $\approx$ 0.88 on 10-fold cross-validation), enabling automated pre-surgical implant identification.

EDUCATION

MSc, Data and Computational Science – University College Dublin (August 2025)
BE, Electronics and Communication Engineering – Kongu Engineering College (May 2020)

CERTIFICATIONS & COURSES

MLOps (Coursera) | Data Science Specialization (Coursera) | Python for ML & Data Science (Udemy) | Snowflake Complete Masterclass (Udemy) | SQL Advanced (Hackerrank) | Microsoft Power BI (Udemy) | Tableau for Data Science (Udemy)